

Titration Calculations

Answer key with worked solutions

A. Titres

1. Definitions.

Titre: volume delivered from the burette. **Aliquot:** an accurately measured portion of a solution (by pipette). **End point:** where the indicator changes colour. **Equivalence point:** where exactly the stoichiometric amount has been added. **Concordant titres:** accurate titres that agree closely (within 0.10 mL).

2. Hydrochloric acid.

(a) Mean of the concordant titres (rough left out):

$$\bar{V} = \frac{(22.60 + 22.70 + 22.65) \text{ mL}}{3} = 22.65 \text{ mL}$$

Deviations -0.05, +0.05, 0.00 mL: $s = \sqrt{\frac{0.0050 \text{ mL}^2}{2}} = 0.05 \text{ mL}$

(b) $n(\text{NaOH}) = 0.1050 \text{ mol/L} \times 0.02265 \text{ L} = 2.378 \times 10^{-3} \text{ mol} = n(\text{HCl})$

$$c(\text{HCl}) = \frac{2.378 \times 10^{-3} \text{ mol}}{0.02500 \text{ L}} = 0.09513 \text{ mol/L}$$

B. Standardisation

3. Sodium carbonate.

(a) $n = \frac{1.325 \text{ g}}{105.99 \text{ g/mol}} = 0.01250 \text{ mol}$; $c = \frac{0.01250 \text{ mol}}{0.2500 \text{ L}} = 0.05000 \text{ mol/L}$

(b) Aliquot: $0.05000 \text{ mol/L} \times 0.02500 \text{ L} = 1.250 \times 10^{-3} \text{ mol Na}_2\text{CO}_3$; $\text{HCl} = 2 \times 1.250 \times 10^{-3} = 2.500 \times 10^{-3} \text{ mol}$

$$c(\text{HCl}) = \frac{2.500 \times 10^{-3} \text{ mol}}{0.02480 \text{ L}} = 0.1008 \text{ mol/L}$$

4. KHP.

$$n = \frac{0.5105 \text{ g}}{204.22 \text{ g/mol}} = 2.500 \times 10^{-3} \text{ mol} = n(\text{NaOH}); c = \frac{2.500 \times 10^{-3} \text{ mol}}{0.02495 \text{ L}} = \mathbf{0.1002 \text{ mol/L}}$$

C. Diprotic acids and dilutions

5. Sulfuric acid.

$$n(\text{NaOH}) = 0.2000 \text{ mol/L} \times 0.01825 \text{ L} = 3.650 \times 10^{-3} \text{ mol}; n(\text{H}_2\text{SO}_4) = \frac{1}{2} \times 3.650 \times 10^{-3} = 1.825 \times 10^{-3} \text{ mol}; c = \frac{1.825 \times 10^{-3} \text{ mol}}{0.02500 \text{ L}} = \mathbf{0.07300 \text{ mol/L}}$$

6. Vinegar.

$$\begin{aligned} \text{Aliquot: } & 0.1000 \text{ mol/L} \times 0.02330 \text{ L} = 2.330 \times 10^{-3} \text{ mol} \\ \text{Flask: } & \frac{250.0 \text{ mL}}{25.00 \text{ mL}} = 10 \text{ aliquots, so } 2.330 \times 10^{-2} \text{ mol, all from the 20.00 mL of vinegar.} \\ c = & \frac{2.330 \times 10^{-2} \text{ mol}}{0.02000 \text{ L}} = \mathbf{1.165 \text{ mol/L}} \end{aligned}$$

D. Back titration

7. Antacid.

$$\begin{aligned} \text{HCl added: } & 0.5000 \text{ mol/L} \times 0.02500 \text{ L} = 1.250 \times 10^{-2} \text{ mol} \\ \text{HCl left over (= NaOH): } & 0.2500 \text{ mol/L} \times 0.01120 \text{ L} = 2.800 \times 10^{-3} \text{ mol} \\ \text{HCl reacted: } & 1.250 \times 10^{-2} - 2.800 \times 10^{-3} = 9.700 \times 10^{-3} \text{ mol} \\ \text{Mg(OH)}_2 \text{ (1 : 2): } & \frac{1}{2} \times 9.700 \times 10^{-3} = 4.850 \times 10^{-3} \text{ mol; mass} = 4.850 \times 10^{-3} \text{ mol} \times \\ & 58.32 \text{ g/mol} = 0.2829 \text{ g} \\ \text{Percent: } & \frac{0.2829 \text{ g}}{0.300 \text{ g}} \times 100\% = \mathbf{94.3\%} \end{aligned}$$

E. Challenge

8. Uncertainty.

$$\text{Two readings: } 2 \times 0.05 \text{ mL} = 0.10 \text{ mL}; \frac{0.10 \text{ mL}}{22.65 \text{ mL}} \times 100\% = \mathbf{0.44\%}. \text{ To reduce it, make the titre } \mathbf{\text{larger}}$$
 (for example, use a more dilute titrant or a larger aliquot), so the same $\pm 0.10 \text{ mL}$ is a smaller fraction of the titre.